

41891: Cloud Computing Infrastructure
42891: Infrastructure for Cloud Computing

Week 6: Management of Data Center

Contents

- * Background of Data Center
 - * Key benefits
- * Data center design standards
 - * Uptime Institute's Tier Standard
 - * TIA-942 Standard
- * Key management areas
- * Data center management challenges
- * Hardening steps for data center management

Key Benefits

- * Cost and energy saving
- * Reduced server downtime
- * Increased productivity
- * Improved security
- * Compliance
- * Etc.

Data Center Design Standards

- * Uptime Institute's Tier Standard
- * TIA-942 Standards
- * Energy Star (USA)
- * Data Center Star Audit (Germany)

Uptime Institute's Tier Standard

- * Develops performance-based methodology
- * Tier I – Basic capacity:
 - * Provides dedicated site infrastructure to support IT systems
- * Tier II – Redundant capacity components:
 - * Provides maintenance and safety opportunities
- * Tier III – Concurrently maintainable:
 - * Requires NO shut down for maintenance
- * Tier IV – Fault tolerance:
 - * Prevents interruptions

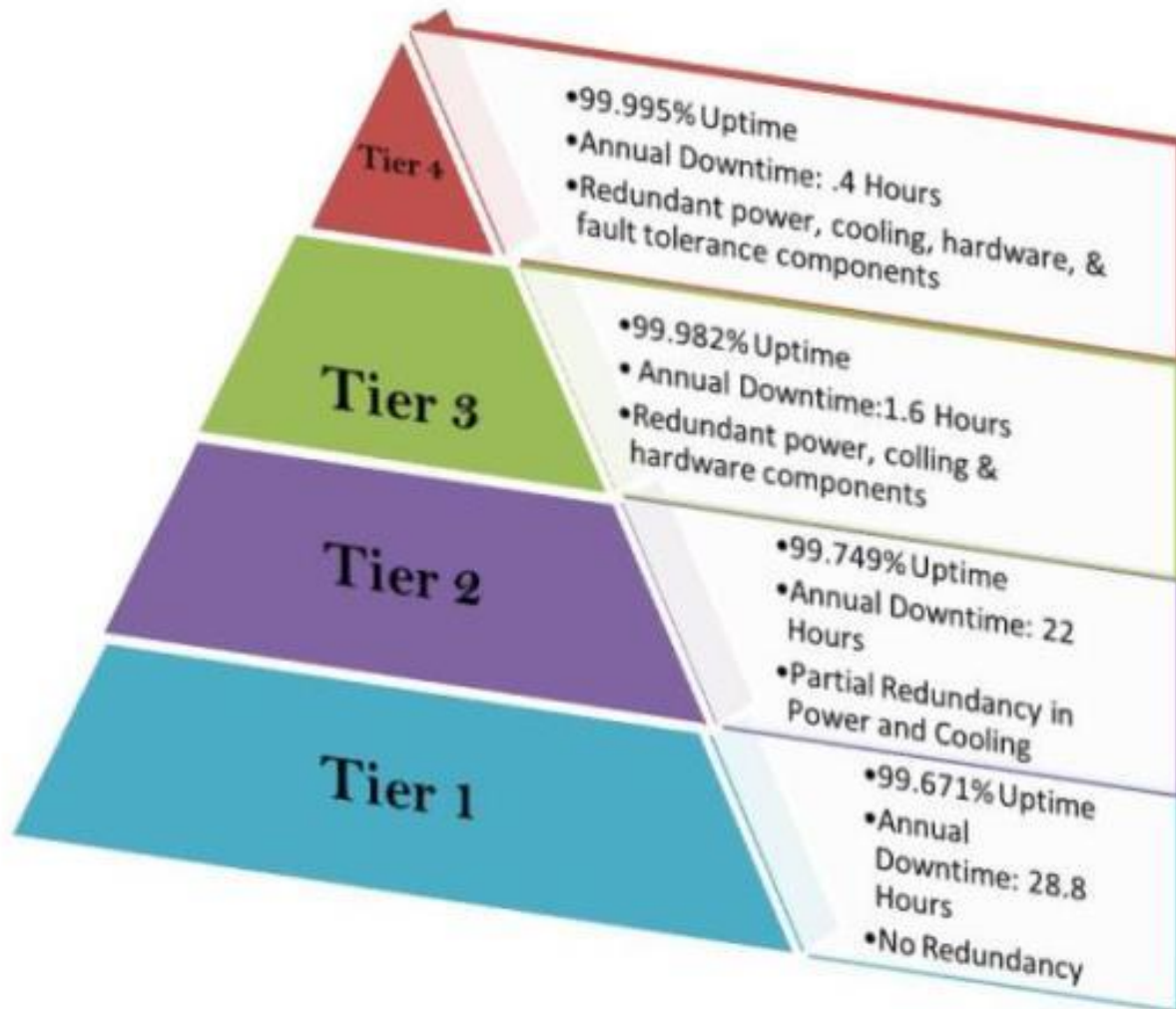
Tier Requirements Summary

	Tier I	Tier II	Tier III	Tier IV
Active Capacity Components to Support the IT Load	N	N+1	N+1	N After any Failure
Distribution Paths	1	1	1 Active and 1 Alternate	2 Simultaneously Active
Concurrently Maintainable	No	No	Yes	Yes
Fault Tolerance	No	No	No	Yes
Compartmentalization	No	No	No	Yes
Continuous Cooling	No	No	No	Yes

TIA-942 Standards

- * Developed by Telecommunications Industry Association (TIA)
- * More of IT cable and network oriented
- * Origins from Uptime Institute's four tiered approach
- * Construction cost varies from tier to tier

TIA-942 Tier Standards



Contd..

- * Tier I – Basic server room:

- * Susceptible to disruptions from planned and unplanned activities
- * Single path for power and cooling distribution
- * No redundant components (N)
- * May or may not have raised floors, UPS or generator
- * Annual downtime of 28.8 hours
- * Complete shut down for maintenance

- * Tier II – redundant components:

- * Less susceptible to disruptions from planned and unplanned activities
- * Single path for power and cooling distribution
- * Includes redundant components (N+1)
- * Includes raised floors, UPS and generator
- * Annual downtime of 22 hours
- * Shut down for power path maintenance

Contd..

- * Tier III – Concurrently maintainable:
 - * Planned activities run without disruptions but unplanned cause disruptions
 - * Multiple power and cooling distribution paths but with only one path active
 - * Includes redundant components (N+1)
 - * Annual downtime of 1.6 hours
 - * Includes raised floors and sufficient capacity and distribution to carry load on one path while performing maintenance on other
- * Tier IV – Fault tolerance:
 - * Planned activity does not disrupt and can handle one worst unplanned event with no critical load impact
 - * Multiple active power and cooling distribution paths
 - * Includes redundant components (2(N+1); i.e. 2 UPS)
 - * Annual downtime of 0.4 hours (24 minutes)

Tier Requirements Summary

Attribute / Statistic	Tier I	Tier II	Tier III	Tier IV
Power and Cooling Delivery Paths	1 Active	1 Active	1 Active 1 Passive	2 Active
Redundant Components	N	N + 1	N + 1	2(N + 1)
Support Space to Raised Floor Ratio	20%	30%	80 – 90%	100%
Initial Watts / sqft	20 – 30	40 – 50	40 – 60	50 – 80
Ultimate Watts / sqft	20 – 30	40 – 50	100 – 150	150+
Raised Floor Height	12"	18"	30 – 36"	30 – 36"
Floor Loading Pounds / sqft	85	100	150	150+
Utility Voltage	208, 480	208, 480	12 – 15 kV	12 – 15 kV
Months to Implement	3	3 – 6	15 – 20	15 – 20
Year First Deployed	1965	1970	1985	1995
Construction \$ / sqft	\$450	\$600	\$900	\$1,100+
Annual IT Downtime Due to Site	28.8 hrs	22.0 hrs	1.6 hrs	0.4 hrs
Site Availability	99.67%	99.75%	99.98%	100.00%

Key Management Areas

- * Capacity Management
- * Service Delivery
- * Resource Management
- * Asset Management

Capacity Management

- * Rack space
 - * Plan from start and use space effectively
- * Network connectivity
 - * Use of DCIM solutions to monitor port connections and connected devices
- * Power
 - * DCIM solution provides granular and holistic power usage information
- * Cooling
 - * Extract infrastructure information and feed it into the computational fluid dynamics (CFD) modelling software tool to understand heating and improve cooling

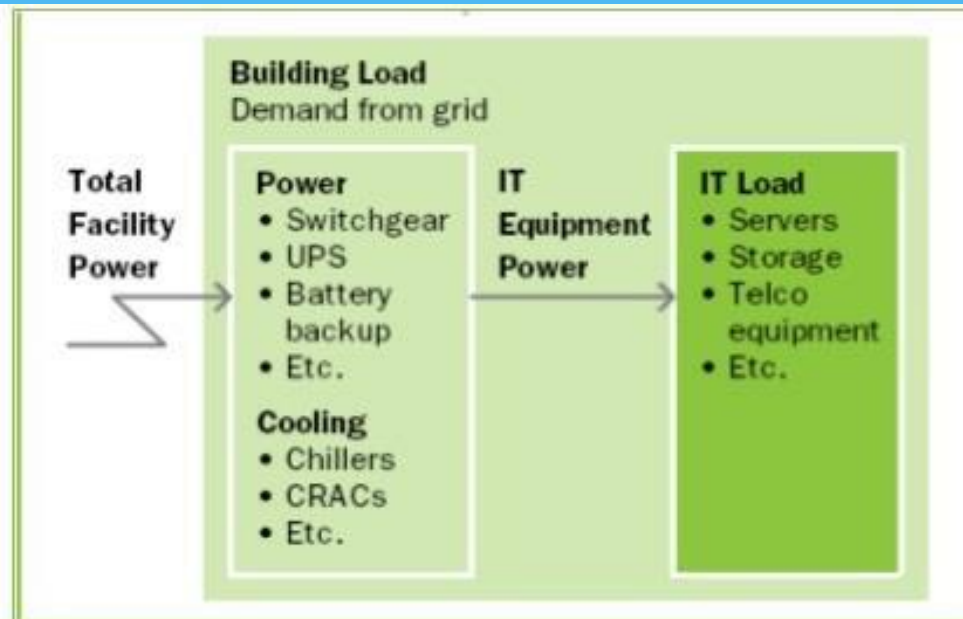
Service Delivery

- * World-class delivery
- * Trust and transparency
- * True multi-tenancy
- * Proven scale
- * High performance
- * Complete disaster recovery
- * High availability

Resource Management

- * Data center optimisation
 - * Cutting electricity bills
 - * Reducing brown energy usage
 - * Scheduling more data centers on Green energy
 - * Minimising response time
- * Temperature-aware resource management
 - * Temperature-reactive
 - * Decisions made on real-time temperature to avoid server overheating
 - * Temperature-proactive
 - * Explicitly take into account the decisions on temperature increase

Power Usage Effectiveness (PUE)



$$\text{PUE} = \text{Power Usage Effectiveness} = \frac{\text{Total Facility Power}}{\text{IT Equipment Power}}$$

$$\text{DCiE} = \text{Data Center Infrastructure Efficiency} = \frac{1}{\text{PUE}} = \frac{\text{IT Equipment Power}}{\text{Total Facility Power}}$$

Contd..

- * Created by members of Green Grid, an industry group
- * Industry standard to determine the energy efficiency/performance
- * Ratio of how efficiently a computer data center uses energy
- * Data center infrastructure efficiency (DCiE) is reciprocal of PUE
- * The Green Grid recently proposed a ranking system for data centres based on their DCiE metric , this ranged from recognised at a DCiE of 0.4 (PUE of 2.5) up to platinum for data centres achieving a DCiE of between 0.8 and 0.9 (PUE 1.25 – 1.11).

Asset Management

- * Knowing assets physical location
- * Report on assets
 - * Physical
 - * Softwares
- * Warranty, contracts and lifetimes of hardware and software assets
- * Automation

Data Center Management Challenges

- * Maintaining availability and uptime
 - * Use a single repository for storing asset information
 - * Enables consistent and accurate record keeping
 - * Reducing troubleshooting time
- * Improving utilization of capacity
 - * Allocate space, manage power and network connectivity
- * Reporting reduced operating expenses
 - * Aggregated data to show stakeholders full operational efficiency of the data center

Contd..

- * Managing energy usage and costs
 - * Monitor energy consumption
 - * Environmental sensors for energy bill back program for energy efficient behaviour
- * Improving staff productivity
 - * Automate manual processes
 - * Assists with provisioning and auditing

Hardening Steps for Data Center Management

- * Environmental control
 - * Temperature and humidity
 - * Fire protection
 - * Flood protection
- * Physical security
 - * Location of data center
 - * Surveillance
 - * Physical access control devices
 - * Assignment of physical access rights

Contd..

- * Disaster recovery
 - * Disaster recovery plan
 - * Offsite backup
- * Remote data center management
 - * Logical security requirement
- * Others
 - * IT staff training
 - * Operational procedures

Summary

- * Need management solutions that operates data center at minimum cost
- * Maximise organisation's capacity to control data center spending
- * Utilise IT assets more effectively
- * Proper planning, competent IT staff, sophisticated management tools
- * Considering environmental, physical, logical and procedural levels to reinforce data center security